SMOOTH, ETALE AND UNRAMIFIED REPRESENTABLE TRANSFORMATIONS ARE FORMALLY SO

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 2528-2538 of sources/stacks-spaces-more-morphisms/pinned-original-source.tex; proof span is lines 2540-2564, the complete author proof (23 body lines).

Editorial note. Navigation stubs: 4 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Representable transformations of functors, smooth/etale/unramified morphisms, and their formal analogues; the reduction is retained.

Source reference. Original lemma 049S: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 2443–2469. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 02HM: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 2564–2564. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 02H6: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 2564–2564. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 02HE: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 2564–2564. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let S be a scheme. Let $F, G : (Sch/S)^{opp}_{fppf} \to Sets$. Let $a : F \to G$ be a representable transformation of functors.*

(1) *If a is smooth then a is formally smooth.*
(2) *If a is étale, then a is formally étale.*
(3) *If a is unramified, then a is formally unramified.*

Date: stacks-project:spaces-more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Proof. Consider a solid commutative diagram

$$\begin{array}{ccc} F & \longleftarrow & T \\ {\scriptstyle a}\downarrow & \nearrow & \downarrow{\scriptstyle i} \\ G & \longleftarrow & T' \end{array}$$

as in Definition 049S. Then $F \times_G T'$ is a scheme smooth (resp. étale, resp. unramified) over T'. Hence by More on Morphisms, Lemma 02H6 (resp. Lemma 02HM, resp. Lemma 02HE) we can fill in (resp. uniquely fill in, resp. fill in at most one way) the dotted arrow in the diagram

$$\begin{array}{ccc} F \times_G T' & \longleftarrow & T \\ \downarrow & \nearrow & \downarrow{\scriptstyle i} \\ T' & \longleftarrow & T' \end{array}$$

an hence we also obtain the corresponding assertion in the first diagram. □